



AI Based Symptom Checker App Using NLP and Machine Learning

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Abstract

In the digital age, the need for quick and reliable access to healthcare information is increasing. This project introduces an AI-powered Symptom Checker Application that leverages Natural Language Processing (NLP) and Machine Learning (ML) to interpret user-described symptoms and diseases. Through NLP processes such as tokenization, lemmatization, and Named Entity Recognition, the application converts free-form text into structured data. This processed information is then fed into ML models like Naïve Bayes, Random Forest, and Support Vector Machine (SVM) to predict possible medical conditions. Acting as a virtual medical assistant, the system offers users an initial assessment of their health concerns, helping them make informed decisions before consulting a doctor. By integrating these technologies, the project enhances healthcare accessibility, supports preventive diagnosis, and reduces misinformation from unreliable sources.

Keywords: Artificial Intelligence (AI), Natural Language Processing (NLP), Machine Learning (ML), Symptom Prediction, Symptom Analysis, Virtual Medical Assistant, Healthcare Automation, Named Entity Recognition (NER), Predictive Modeling, Health Informatics.

1. Introduction

Healthcare systems are rapidly evolving as people increasingly seek digital platforms for medical guidance. Many individuals rely on the internet for health-related information, yet most sources lack medical accuracy. Hence, there is a growing need for an intelligent, automated, and user-friendly system that can analyze symptoms and predict possible diseases based on user input. The proposed AI-Based Symptom Checker aims to meet this demand. It accepts textual symptom descriptions such as "I have a sore throat and mild fever" and processes them using NLP techniques. The extracted symptoms are then matched against a curated dataset, and ML algorithms predict the most probable diseases. This project does not replace professional diagnosis but rather complements it by providing preliminary

their self-reported symptoms, without the need for immediate professional consultation. By acting as a virtual medical assistant, the system provides a level of diagnosis that helps users make informed healthcare decisions. The application integrates Natural Language Processing (NLP) and Machine Learning (ML), which analyze natural language inputs, identify medical terms, and correlate them with patterns stored in a trained dataset. This is achieved through user interaction, where individuals describe symptoms in natural or conversational language. Since these inputs differ in structure from standard NLP data, the system uses Natural Language Processing (NLP) techniques for accurate interpretation. The NLP process involves tokenization, stop-word removal,